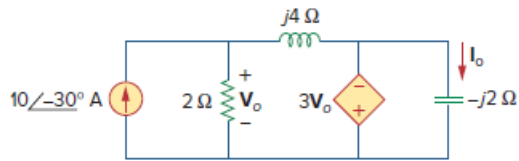


## Problem

Determine  $V_o$  and  $I_o$  in the circuit of Fig. 10.80 using mesh analysis.

Figure 10.80:



## Step-by-step solution

## Step 1 of 4

Refer to Figure 10.80 in the textbook.

Draw the circuit diagram with mesh currents.

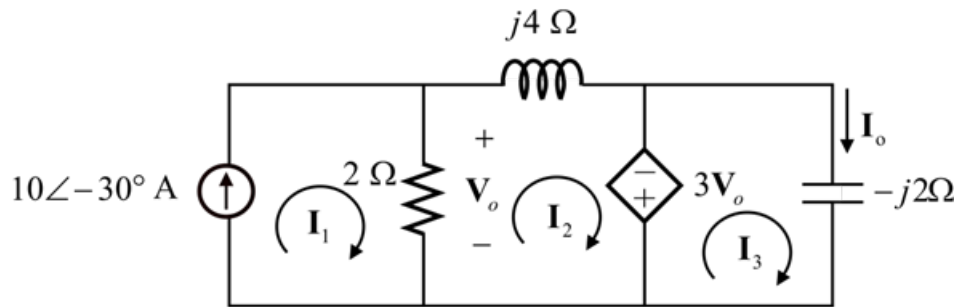


Figure 1

## Step 2 of 4

From Figure 1, the value of current,  $I_1$  is,

$$I_1 = 10\angle -30^\circ \text{ A}$$

The value of voltage,  $V_o$  is,

$$V_o = 2(I_1 - I_2)$$

$$I_1 - I_2 = 0.5V_o$$

$$\begin{aligned} I_2 &= I_1 - 0.5V_o \\ &= 10\angle -30^\circ - 0.5V_o \end{aligned}$$

## Step 3 of 4

Apply mesh analysis for loop 2 to calculate the output voltage.

$$-V_o + (j4)I_2 - 3V_o = 0$$

Substitute  $10\angle -30^\circ - 0.5V_o$  for  $I_2$ .

$$-4V_o + (j4)[10\angle -30^\circ - 0.5V_o] = 0$$

$$(4 + j2)V_o = 40\angle 60^\circ$$

$$V_o = 8.944\angle 33.435^\circ \text{ V}$$

Therefore, the value of voltage,  $V_o$  is  $8.944\angle 33.435^\circ \text{ V}$ .

#### Step 4 of 4

Apply mesh analysis for loop 2 to calculate the output current.

$$3V_o - (j2)I_o = 0$$

$$I_o = \frac{3(8.944\angle 33.435^\circ)}{j2}$$

$$I_o = 13.416\angle -56.565^\circ \text{ A}$$

Therefore, the value of current,  $I_o$  is  $13.416\angle -56.565^\circ \text{ A}$ .

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[Comments \(1\)](#)

☐

**Anonymous**

it should say Loop 3

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